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Is It the Government's Responsibility to Reduce Income Inequality? An Age-Period-Cohort Analysis of Public Opinion toward Redistributive Policy in the United States, 1978 to 2016

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ABSTRACT

This study examines age, period, and cohort (APC) effects on changing opinions among the American public toward the federal government's responsibility for income redistribution. More specifically, we use the hierarchical age–period–cohort (HAPC) model to analyze time periods and birth cohorts as contextual variables and age as an individual-level variable to address the identification problem inherent in APC analysis. Our results show that while cohort effects on public opinion toward redistributive policy exist, such effects are explained by individual-level compositional differences among cohorts, and while period effects are evident, particular trends in that regard are difficult to discern. What our results make abundantly clear, however, is the significant impact of age on opinions toward redistributive policy. As people age, they become significantly less supportive of federal redistribution policies, a relationship that is robust in the presence of cohort and period effects as well as a full range of controls. In the context of a rapidly aging population, the implication is that more conservative policy preferences linked to older age provide little reason to believe that mass support for government redistribution is in the offing.

Introduction

Public opinion in the United States about the proper role of government in terms of redistributive policy has been subject to change over time, due in part to flux in the social, economic, and political environment (Schneider and Jacoby 2005). For example, the New Deal coalition that emerged from the economic crisis of the Great Depression in the 1930s and held sway in American politics through the 1960s greatly expanded the scope of the federal government and its role in redistributive policies. In just 40 years the New Dealers created many of the programs that are now cornerstones of the U.S. welfare state, including social security, Medicare, Medicaid, food stamps (now the Supplemental Nutrition Assistance Program), and Aid to Families with Dependent Children (now Temporary Assistance to Needy Families). But beginning in the 1970s the national mood began to shift and, with President Reagan being emblematic, the United States entered a new political period characterized by calls for less government intervention in the economy and less spending on antipoverty policy in particular (Iceland 2013).

Important to note, the post-WWII U.S. economy has also been characterized by two qualitatively different periods. From 1945 through the 1960s median incomes rose and income

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inequality decreased. Then, beginning in the 1970s, in what has been coined the “Great U-Turn,” average incomes began to stagnate, and income inequality began a steady upward march that continues to this day (Harrison and Bluestone 1988; Piketty and Saez 2003).

Despite increased popular recognition of rising income inequality, as demonstrated recently by the Occupy Wall Street Movement, large-scale shifts in U.S. public opinion toward redistributive policies have not followed suit. In among the most comprehensive treatments of the subject to date, McCall (2013) showed that despite Americans becoming increasingly concerned about inequality in recent decades, this worry has not translated into greater public support for traditional redistributive programs (see also Kenworthy and McCall 2007; Kenworthy and Pontusson 2005). In part, McCall (2013) attributed this to the simultaneous and offsetting strength of antitax attitudes and negative sentiments toward social welfare spending. In fact, not only is there limited evidence that inequality generates greater mass preference for government action, but some research has actually shown the opposite: As inequality rises there is a general public shift toward more *conservative* political attitudes (Benabou 2000; Kelly and Enns 2010; Luttig 2013; Trump 2017; Wright 2017). While there is no consensus on this relationship (e.g., Johnston and Newman 2016), it is a finding that parallels the different aforementioned political and economic moments in the post-WWII era. Other studies do not stress trends in inequality as the major drivers in opinions toward redistributive programs in recent decades, but instead social affinities and cleavages in the American electorate related to compositional factors, such as religious affiliation (Scheve and Stasavage 2006); growing racial and ethnic diversity (Alesina and Glaeser 2004); and, important to note for our purpose, age (Alwin 2002). In fact, the effects of age on public attitudes toward social programs remain a somewhat open question. For instance, in some studies older people are shown to hold less supportive opinions regarding redistributive policies (Brooks and Manza 2013; Hayes 2011), while others have found no significant relationship between age and policy attitudes (Steele 2015). Gaining a better understanding of the influence of age on policy positions takes on special significance when considered in light of trends in population aging. Those ages 65 and older are projected to more than double in number from 46 million today to more than 98 million by 2060, a corresponding rise from 15% to 24% of the total U.S. population (Mather, Jacobsen, and Pollard 2015).

In this article we examine how public sentiment toward federal income redistribution has changed over the last four decades by disentangling the joint influences of age group, time period, and birth cohort on political opinions. Doing so makes a unique contribution to the literature in two ways. First, although age–period–cohort (APC) models have figured prominently in other types of studies of social change, the approach taken here has not previously been applied to understanding the politics of redistribution. Second, isolating age effects using this method takes on increasing substantive importance in the context of a rapidly aging U.S. electorate.

Disentangling age, period, and cohort effects

The examination of APC effects are central concepts in the demographic study of social change. Age effects refer to the ways in which people change as they progress through their lives from being younger to older. Period effects refer to the social, economic, and political context that characterizes a particular point in time. Cohorts are social groups with a distinctive composition and character owing to their shared origin and history (Ryder 1965). The most common example is the birth cohort, a group of people united by birth year who pass through history together at the same stage in their life cycle. Cohort effects are generally attributed to size-based or history-based effects. Size-based effects posit that the size of a cohort has important influences on social outcomes. For example, Easterlin (1980) argued that large birth cohorts were disadvantaged relative to small birth cohorts because larger group numbers lead to greater competition for scarce societal resources, while lower group numbers mean relative resource abundance. History-based

effects, on the other hand, are thought to be imbedded in a cohort due to its shared history, creating lasting impacts on collective identity, beliefs, and behaviors. For instance, the influence of the Great Depression and World War II created the character of those who have been called the “Greatest Generation” (Brokaw 1997). Or due to the enduring impact of disenfranchisement on women who came of age before the 19th Amendment, the cohort remained much less likely to vote even after women’s suffrage became a constitutional right (Firebaugh and Chen 1995).

When it comes to political ideology in the United States, it is popular wisdom that people grow more conservative as they age. However, a series of recent studies by Gallup (Newport, Jones, and Saad 2014) and the Pew Research Center (2011; Desilver 2014) suggest the relationship may not be so clear once generational differences are accounted for. Research suggests that the political environment in which generations come of age has lasting consequences over the life course. For example, Pew Research Center (2011) shows that the relative popularity of the two major political parties and the president when a person reaches voting age (i.e., 18 years old) carries lasting consequences for voting over one’s lifetime. Moreover, research by Gallup (Newport et al. 2014) shows that the baby boomers—the large generation of Americans born between 1946 and 1964 and a major political force due to their relative size—tend to lean more Republican than the general electorate. However, there are also important variations among the baby boomers that may be attributable to the period within which they came of voting age. For example, the most conservative of the baby boomers are those born after 1960 who reached voting age during the stagflation and tax revolts of the 1970s, tense Cold War relations, and President Ronald Reagan’s election in 1980. In contrast, the millennials born since 1980 tend to be disproportionately liberal (Desilver 2014). This has been attributed in part to the rise of the Internet and shifting social views on subjects such as same-sex marriage, as well as the fact that millennials are the most racially and ethnically diverse generation in U.S. history and non-whites are disproportionately supportive of Democrats (Pew Research Center 2011). All of this said, if age remains a driving force related to policy attitudes, net of period and cohort, this holds tremendous implications given that “the current growth of the population 65 and older is one of the most significant demographic trends in the history of the United States” (Mather et al. 2015:2).

A challenge in sussing out the joint effects of APC is the exact linear dependence between the three variables (i.e., cohort = period – age). That is, if the year is 2017 and an individual is 17 years old, you know that person is a member of the 1999–2000 birth cohort. One approach for dealing with this identification problem is the hierarchical age–period–cohort (HAPC) model developed by Yang and Land (2006). The HAPC model draws on hierarchical linear modeling, a form of regression analysis that analyzes variance in the dependent variable while accounting for the hierarchical structure of independent variables. The lower level of analysis is termed the first level of analysis, while the aggregate level of analysis in which first-level variables are nested is referred to as the second level of analysis. The HAPC model takes advantage of the fact that individuals are nested in periods and cohorts. That is, period and cohort effects are treated as second-level variables and their effects are estimated separately from first-level variables, including age, thus eliminating the APC identification problem. In what follows, we use the HAPC model to assess the joint influence of APC effects on public support for federal income redistribution.

Methodology

The data used in the present study are drawn from the General Social Survey (GSS). The GSS has been collected in the United States every year from 1972 to 1993 except 1979, 1981, and 1992, and then every even year since 1994. Because the item we use to operationalize our dependent variable was not asked until 1978, our analysis is restricted to data collected from 1978 through 2016, covering 23 periods over 38 years (this question was not asked in every year during this

time frame). While the original sample size was 32,838, listwise deletion of missing cases reduced the sample to 28,927 cases available for our analysis.¹

Dependent variable

The question *eqwlth* in the GSS reads as follows:

Some people think the government in Washington ought to reduce the income differences between the rich and the poor perhaps by raising the taxes of wealthy families or giving income assistance to the poor. Others think that the government should not concern itself with reducing this income difference between the rich and the poor. Here is a card with a scale from 1 to 7. Think of a score of 1 as meaning that the government ought to reduce the income differences between rich and poor, and a score of 7 meaning that the government should not concern itself with reducing income differences. What score between 1 and 7 comes closest to the way you feel?

We reverse-coded the variable so that higher scores indicate more progressive opinions toward federal income redistribution.

Independent variables

The key independent variables in our analysis are age, period, and cohort. Age is a continuous measure in years among those age 18 and older. Period is a categorical measure capturing the survey year. We have 23 periods: 1978, 1980, 1983, 1984, 1986–1991, 1993, 1994, 1996, 1998, 2000, 2002, 2004, 2006, 2008, 2010, 2012, 2014, and 2016. Cohort is measured as 10-year birth cohorts for those born between 1880 and 1990. For example, those who were born in 1985 would be counted in the 1980s cohort. Thus, there are 12 cohorts in our sample. Finally, we control for a range of individual-level factors, including sex, race/ethnicity, education, marital status, household income, religion, and region.

Analytic strategy

To run the HAPC model, it is necessary to establish a cross table including information on periods and cohorts. Table 1 shows the number of respondents by period and cohort. Based on the table, we use hierarchical linear modeling to analyze the aggregate effects of periods and cohorts on individual-level opinions toward federal redistributive policy. Following the principle advocated by Raudenbush and Bryk (2002), we first present an unconditional model without any control variables and then add individual-level control variables (conditional model).

Our model is specified as follows:

Individual Level: Opinion toward government's role_{ijk} = $\beta_{0jk} + \beta_1(\text{Female})_{ijk} + \beta_2(\text{Age})_{ijk} + \beta_3(\text{Education})_{ijk} + \beta_4(\text{Marital status})_{ijk} + \beta_5(\text{Income})_{ijk} + \beta_6(\text{Black})_{ijk} + \beta_7(\text{Other Races})_{ijk} + \beta_8(\text{Catholic})_{ijk} + \beta_9(\text{Other Religions})_{ijk} + \beta_{10}(\text{No Religion})_{ijk} + \beta_{11}(\text{Midwest})_{ijk} + \beta_{12}(\text{South})_{ijk} + \beta_{13}(\text{West})_{ijk} + \varepsilon_{ijk}$

Second Level: $\beta_{0jk} = \gamma_0 + \mu_{0j} + \nu_{0k}$

where β_{0jk} is the intercept at the individual level, indicating the average score in the *j* cohort and the *k* period; γ_0 is the intercept for second level. It represents the average score for all respondents; μ_{0j} is the residual for the *j* cohort in the random effects model; and ν_{0k} is the residual for the *k* period in random effects model.

¹Approximately 95% of our missing cases were due to the household income variable, which we treat as a control. Since there is no evidence of systematic missingness on our dependent variable or key independent variables, we elected to rely on *available case analysis*, which is advocated as the best approach for dealing with missing data in regression analysis when statistical power is not an issue (Howell 2008).

Table 1. Number of respondents by cohort and period.

Birth cohort	Period																							
	1978	1980	1983	1984	1986	1987	1988	1989	1990	1991	1993	1994	1996	1998	2000	2002	2004	2006	2008	2010	2012	2014	2016	Total
1880	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3
1890	28	25	13	6	5	6	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	88
1900	72	79	69	61	62	55	24	32	19	19	13	19	14	3	0	0	0	0	0	0	0	0	0	541
1910	83	176	149	139	125	146	94	70	67	66	65	86	53	64	44	23	4	18	12	0	0	0	0	1484
1920	102	191	191	168	166	196	113	112	98	95	90	170	129	138	117	47	42	82	43	45	36	27	14	2412
1930	108	183	189	162	162	204	99	118	76	105	102	197	158	140	135	75	58	126	92	81	80	89	91	2830
1940	160	297	277	271	259	313	161	156	139	140	160	306	288	234	238	110	111	209	151	147	154	172	170	4623
1950	142	314	418	329	355	425	196	225	200	253	252	432	383	378	348	182	145	341	247	241	194	279	322	6601
1960	0	51	112	181	190	298	199	195	158	186	197	438	402	401	371	179	172	356	227	220	220	269	318	5340
1970	0	0	0	0	0	0	2	18	17	36	60	115	226	291	302	161	157	341	225	214	221	298	304	2988
1980	0	0	0	0	0	0	0	0	0	0	0	0	0	2	51	39	76	217	190	218	217	299	316	1625
1990	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	36	57	84	211	392
Total	698	1316	1418	1317	1324	1643	893	926	774	900	939	1763	1653	1651	1606	816	765	1690	1191	1202	1179	1517	1746	28927

Table 2. Descriptive statistics.

	Range	M (SD)
Opinion toward government's role	1–7	4.3 (1.97)
Age (in years)	18–89	45.4 (17.0)
Education (in years)	0–20	13.1 (3.1)
Household income (in \$1,000)	0.4–178.7	45.6 (37.4)
	Count	%
Gender		
Male	13,056	45.1
Female	15,871	54.9
Marital status		
Married	14,772	51.1
Single	14,155	48.9
Race		
White	23,283	80.5
Black	3,954	13.7
Other races	1,690	5.8
Religion		
Protestant	16,399	56.7
Catholic	7,004	24.2
Other religions	1,927	6.7
No religion	3,597	12.4
Region		
Northeast	5,269	18.2
Midwest	7,426	25.8
South	10,130	35.1
West	6,102	21.1

Note: $N = 28,927$.

Results

Table 2 shows descriptive statistics for the variables included in our analysis. The mean value of public opinion toward the federal government's role in income inequality is 4.3, leaning slightly in the liberal direction (the midpoint is 4.0). The average age is 45.4 years, the mean year of education is 13.1, and the average household income (adjusted for inflation, in 2000 dollars) is \$45,555. The sample includes more females than males (54.9% vs. 45.1%) and is predominately white (80.5%). Protestant (56.7%) and Catholic (24.2%) are the two major religious affiliations in our sample. The largest share of respondents are located in South (35.1%), followed by Midwest (25.7%), West (21.1%), and Northeast (18.2%).

In Figure 1, we can see how the opinions of Americans regarding the federal government's responsibility for reducing income inequality changed during the survey period. The average scores have been above 4.0 every year, leaning toward the opinion that the federal government should engage in redistributive policy. However, there is also evidence of more conservative opinions toward redistribution policy in some periods, including 1980, the mid- to late 1990s, and 2010.

Figure 2, on the other hand, shows the mean public opinion scores toward redistributive policy by cohort. From the 1880s cohort to the 1930s cohort the scores gradually decreased, indicating lessening support for government redistribution among the early cohorts in our sample. After the 1930s cohort, however, the support for redistributive policy steadily increases, with the youngest cohort (born in the 1990s) showing the most positive opinion toward the federal government's responsibility to redistribute income.

Figure 3 shows mean public opinion scores toward redistributive policy by age. The results show that as people grow older, they increasingly disagree with the idea that it is the federal government's responsibility to redistribute income. This clearly fits with popular notions of the relationship between age and trends in people's political views.

Table 3 shows the results from the random effects HAPC model. Model 1 is the unconditional model in which we consider only the effects of APC. The model confirms that people become

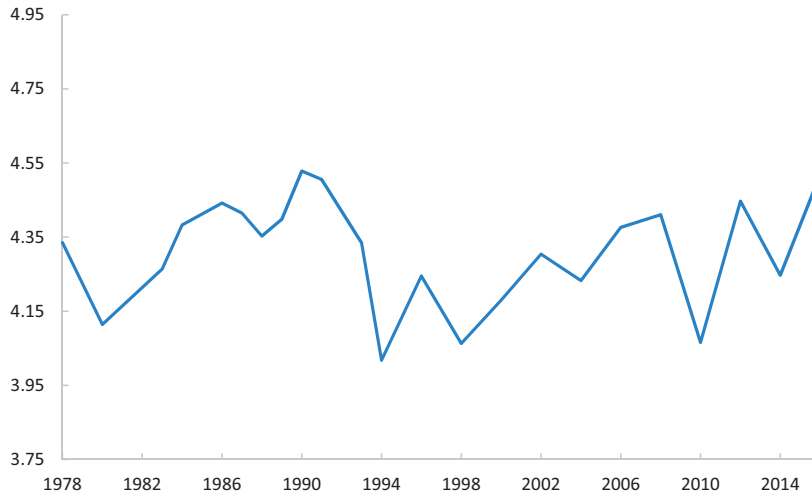


Figure 1. Mean values of public opinion toward redistributive policy by period. *Note.* Higher scores indicate greater preference for government redistribution.

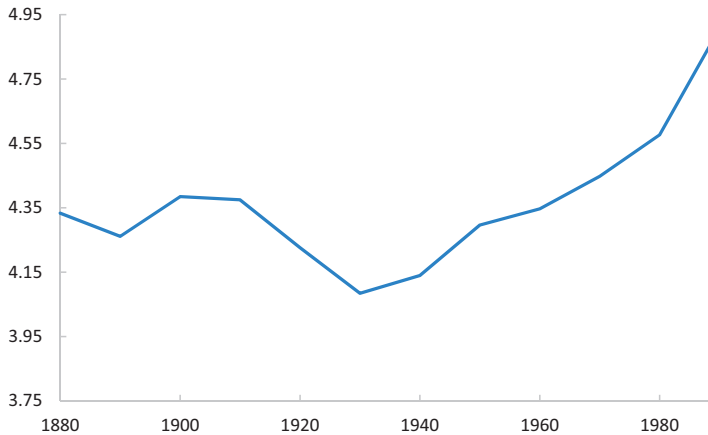


Figure 2. Mean values of public opinion toward redistributive policy by cohort. *Note.* Higher scores indicate greater preference for government redistribution.

more conservative with age. The results show that people born between 1900 and 1919, who would have come of age during the Great Depression, are more supportive of progressive redistribution policy. Meanwhile, people born from 1930 to 1959 show significantly more conservative opinions, before more progressive views reemerge in the 1990s cohort. The model also shows significant period effects, though clear trends in this regard are difficult to discern.

Model 2 shows the conditional model including all control variables.² In the presence of individual-level controls, none of the cohort effects reaches statistical significance. This suggests that the cohort effects observed in Model 1 are not independent but instead are explained by compositional differences among cohorts, such as age, educational attainment, and race/ethnicity.

²In the review process we were asked to include party affiliation in our models. Ancillary analysis showed that the inclusion of this variable did not significantly alter the results of our model. We thus opted to omit this variable from the final analysis presented here.

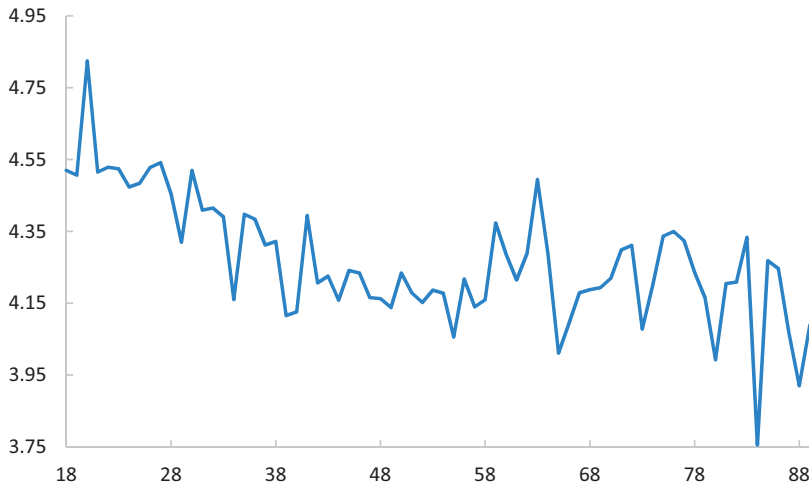


Figure 3. Mean values of public opinion toward redistributive policy by age. *Note.* Higher scores indicate greater preference for government redistribution.

Most of the period effects hold in Model 2, suggesting meaningful impacts related to the economic and political context during particular points in time. That said, particular trends in period effects on public opinion toward federal redistributive policy remain difficult to distill. There is less support for federal redistribution when President Reagan enters office, more support in the latter part of President G. H. Bush's term, less support under President Clinton, and a mixed bag for President Obama.

In terms of APC effects on public opinion toward redistributive policy, Model 2 makes one finding abundantly clear: Age is negatively related to support for the federal government taking an active role in income redistribution. The older people are, the less supportive they are of redistributive policy, even in the presence of the full range of controls.

As for the control variables, Model 2 shows women, younger people, those with low education and low income, and African Americans are more supportive of government redistribution. Protestants are less supportive of progressive redistribution policy compared to people with other religious affiliations. People living in Northeast show more positive opinions toward government income redistribution, compared to people living in South, West, and Midwest.

Discussion and conclusions

In this study we examined change in U.S. public opinion toward federal income redistribution from 1978 to 2016, paying special attention to APC effects. More specifically, we used the HAPC model to analyze time periods and birth cohorts as contextual variables and age as an individual-level variable, thus addressing the identification problem inherent in APC models (i.e., cohort = period – age). Our results showed that while cohort effects on public opinion toward federal redistributive policy exist, such effects are explained by individual-level compositional differences among cohorts. While period effects were evident, particular trends in that regard were difficult to discern. Our results made abundantly clear, however, the significant impact of age on opinions about redistributive policy. As people age, they become significantly less supportive of federal redistribution policies, a relationship that holds in the presence of cohort and period effects as well as a full range of controls.

Our results contribute to the broader literature on U.S. public opinion toward redistributive policy (e.g., Benabou 2000; Kelly and Enns 2010; McCall 2013; McCall et al. 2017). We provided a novel application of the HAPC model to disentangle the joint, and often confounding,

Table 3. HAPC models of opinions toward federal redistributive policy.

Fixed effects	Model 1: Unconditional model		Model 2: Conditional model	
	Coefficient	SE	Coefficient	SE
Intercept	4.389***	0.056	4.310***	0.031
Female			0.292***	0.022
Age ^a	−0.663***	0.146	−0.607***	0.075
Education (in years)			−0.061***	0.004
Married			−0.008	0.024
Household income (in \$1,000)			−0.008***	0.000
Race				
White			—	—
Black			0.779***	0.034
Other			0.305***	0.046
Religion				
Protestant			—	—
Catholic			0.129***	0.028
Other religions			0.305***	0.046
Non religion			0.388***	0.036
Region				
Northeast			—	—
Midwest			−0.127**	0.034
South			−0.326***	0.033
West			−0.288***	0.036
Random effects				
Cohort				
1880–1889	0.004	0.144	0.000	0.025
1890–1999	0.056	0.120	−0.001	0.025
1900–1909	0.163*	0.076	0.001	0.024
1910–1919	0.156**	0.053	0.010	0.023
1920–1929	−0.007	0.045	0.002	0.022
1930–1939	−0.189***	0.043	−0.030	0.021
1940–1949	−0.200***	0.037	0.005	0.020
1950–1959	−0.109**	0.034	0.022	0.018
1960–1969	−0.097**	0.036	−0.004	0.019
1970–1979	−0.010	0.043	−0.001	0.021
1980–1989	0.038	0.053	−0.011	0.023
1990–1999	0.197**	0.085	0.009	0.025
Period				
1978	0.007	0.068	0.005	0.063
1980	−0.185***	0.055	−0.174***	0.049
1983	−0.051	0.054	−0.069	0.047
1984	0.053	0.055	0.011	0.049
1986	0.113*	0.055	0.068	0.049
1987	0.094	0.051	−0.059	0.044
1988	0.035	0.063	0.004	0.057
1989	0.076	0.062	0.074	0.056
1990	0.177**	0.066	0.189**	0.060
1991	0.164**	0.063	0.141*	0.057
1993	0.031	0.062	0.055	0.056
1994	−0.247***	0.050	−0.228***	0.043
1996	−0.050	0.051	−0.029	0.044
1998	−0.215***	0.051	−0.231***	0.044
2000	−0.110*	0.052	−0.081	0.045
2002	0.003	0.065	0.023	0.059
2004	−0.060	0.066	0.027	0.061
2006	0.062	0.051	0.082	0.044
2008	0.092	0.057	0.130*	0.051
2010	−0.209***	0.057	−0.217***	0.051
2012	0.116*	0.057	0.117*	0.051
2014	−0.059	0.053	−0.029	0.046
2016	0.165**	0.051	0.192***	0.043
	Variance		Variance	
Cohort effect var (b_{00j}) = τ_{b00}	0.021***		.0006	
Period effect var (c_{00k}) = τ_{c00}	0.019***		.013***	
Individual effect var (e_{ij}) = σ^2	3.821		3.5091111	
Deviance	120937.432		11845213.409111111111	
df	5		17	

Note: $N = 28,927$. var = variance.^aAge was divided by 100 to show its coefficients.* $p < .05$. ** $p < .01$. *** $p < .001$.

influences of age group, time period, and birth cohort (Yang and Land 2006) to understand the politics of redistribution. While APC models have figured prominently in other types of studies of social change, the approach taken here has not before been applied to this topic. Despite literature noting distinctly different political leanings across U.S. generations (e.g., Desilver 2014; Newport et al. 2014)—for example, contrasting the voting patterns of baby boomers (more conservative) and millennials (more liberal)—our research suggests that in terms of opinions toward federal redistribution, age and other compositional factors are much more salient than is cohort membership. This finding echoes previous public opinion research (e.g., Alesina and Glaeser 2004; Alwin 2002; Caren, Ghoshal, and Ribas 2011; Scheve and Stasavage 2006; Schwadel 2010; Schwadel and Garneau 2014), and takes on special significance given projections that nearly one in four Americans will be older than age 65 by the middle part of this century (Mather et al. 2015). Despite rising income inequality, more conservative policy preferences linked to older age in the context of a rapidly aging population provide little reason to believe that mass support for government redistribution is in the offing.

While making the aforementioned contributions, we must also note limitations of our study, issues that might be attended to in future research efforts. For example, we used only one item to measure public opinion toward the federal government's proper role in income redistribution. It might be more appropriate to operationalize this political preference as a latent variable and use a series of questions to create and validate a scaled measure. Further, while we showed that some period effects matter in shaping public opinion, our models make the answer as to *how* and *why* unclear, especially in the absence of discernable trends over time. Future research efforts using specific period characteristics that more concretely capture elements of the socioeconomic or historical context of a given time period might be helpful to tease these issues out.

In conclusion, we return to Ryder's (1965) early call for understanding the joint influence of APC in the study of social change. The annual infusion of birth cohorts into society represents a critical vehicle for societal transformation as generations pass through the life cycle together. Attention to techniques that allow for the independent identification of APC effects on public opinion and other important outcomes continues to be crucial more than 50 years after Ryder made his original case.

Author notes

Ya-Feng Lin has a PhD from the Department of Sociology, Louisiana State University. His research topics include marriage and family, inequality, demography, quantitative analyses, and comparative studies.

Yoshinori Kamo is Professor and Chair of the Department of Sociology at Louisiana State University. His research interests include family, inequality, psychological well-being, and quantitative methods.

Tim Slack is Professor of Sociology at Louisiana State University. His interests are in the areas of social demography, inequality and poverty, and community and environment.

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